NASSAU CANDY DISTRIBUTOR

Executive Summary

Factory Reallocation & Shipping Optimization Recommendation System

Project Type: Data Analytics, Machine Learning, and Decision Intelligence
Purpose: Improve shipping efficiency, evaluate factory-reallocation alternatives, and support data-driven operational decisions.

---

1. Project Overview

The Nassau Candy Factory Reallocation & Shipping Optimization Recommendation System analyzes historical order and shipping data to evaluate shipping performance, compare predictive models, and identify potential factory-reallocation opportunities.

The project integrates data analysis, machine learning, scenario simulation, and an interactive Streamlit dashboard. It is designed to help business stakeholders explore shipping patterns, evaluate alternative factory assignments, and understand the potential operational and financial implications of their decisions.

The dataset contains **10,194 historical records and 18 original columns**, including order dates, ship dates, shipping modes, product details, sales, units, gross profit, and cost.

2. Key Business Findings

The analysis identifies several areas requiring attention:

- **Shipping-mode distribution:** Standard Class accounts for approximately 60% of records, making it the largest shipping category.
- **Data reliability:** Recorded order and ship dates produce unusually long shipping intervals. These require investigation before the calculated lead times can be treated as operationally accurate.
- **Predictive performance:** The evaluated models do not demonstrate satisfactory predictive performance in the reported results.
- **Factory reallocation:** The simulation identifies alternative factory candidates for evaluation, but these candidates should not be treated as approved or optimal assignments.
- **Operational decision support:** The dashboard provides a foundation for exploring historical performance, comparing scenarios, and identifying areas for further investigation.

3. Shipping-Mode Distribution

The dataset contains the following shipping-mode distribution:

Shipping Mode	Records	Percentage
Standard Class	6,120	60.0%
Second Class	1,979	19.4%
First Class	1,548	15.2%
Same Day	547	5.4%
Total	**10,194**	**100.0%**

Standard Class represents the majority of recorded shipments. This makes it an important category for further analysis, particularly when assessing shipping costs, delivery performance, and opportunities to improve service levels.

However, shipping-mode frequency alone does not establish which mode is most profitable or efficient. Such conclusions require validated delivery intervals, transportation costs, and service-level requirements.

4. Machine-Learning Model Performance

Three models were evaluated for shipping lead-time prediction.

Model	MAE (Days)	RMSE (Days)	R^2
Gradient Boosting	176.33	182.50	-0.0089
Random Forest	177.24	186.03	-0.0484
Linear Regression	180.36	183.05	-0.0150

Interpretation

- Gradient Boosting has the lowest reported MAE and RMSE among the three models.
- All three models have negative R^2 values.
- Negative R^2 indicates that the models perform worse than predicting the mean target value on the evaluated test data.
- The reported errors are very large for a practical shipping lead-time prediction system.

Business conclusion: Gradient Boosting is the strongest of these three models by the reported MAE and RMSE, but none should be considered ready for operational deployment. The date anomalies and model-validation methodology must be investigated before selecting a production model.

5. Data-Quality Risks

A major concern is the relationship between the recorded order dates and ship dates. The available records produce shipping intervals that are unusually long for a conventional distributor shipping dataset.

Potential explanations include incorrect source dates, inconsistent date formats, data-generation issues, or a mismatch between the meaning of the recorded dates and the intended shipping lead-time definition.

Recommended corrective actions include:

1. Confirm the original source and meaning of the order and ship dates.
2. Parse dates explicitly using the correct day-first format where applicable.
3. Investigate unusually long intervals and verify them against the source system.
4. Recalculate shipping lead times only after the date fields have been validated.
5. Retrain and reevaluate models using corrected target values.
6. Compare results with a baseline model using a chronological test set.

Until these checks are completed, lead-time estimates should be clearly labelled as unvalidated.

6. Factory-Reallocation Simulation

The factory-reallocation component evaluates alternative factory assignments for products and regions.

The simulation is intended to help stakeholders investigate whether an alternative factory could offer operational benefits. However, a simulated alternative is not automatically a recommendation.

Before an alternative factory can be approved, the analysis must incorporate:

- Actual factory production capacity and current utilization.
- Product-factory compatibility.
- Manufacturing and handling costs.
- Transportation routes, distances, and freight costs.
- Expected production and delivery times.
- Inventory availability and regional demand.
- Profitability and service-level constraints.

The business should compare each proposed assignment with the current baseline using verified costs, realistic capacity limits, and validated delivery-time estimates.

7. Principal Risks

Risk	Business Impact	Recommended Mitigation
Incorrect or inconsistent dates	Misleading lead times and predictions	Validate dates against original records
Weak predictive performance	Unreliable operational forecasts	Rebuild features, retrain models, and compare against a baseline
Unverified factory capacity	Infeasible production assignments	Obtain capacity and utilization data
Missing route and freight costs	Inaccurate savings estimates	Validate transport routes and cost data
Candidate assignments mistaken for recommendations	Poor allocation decisions	Clearly distinguish simulated candidates from approved recommendations
Incomplete operational validation	Premature deployment	Conduct stakeholder review and a controlled pilot

8. Recommendations

The following actions are recommended, in priority order:

Priority 1 — Validate the data

Resolve the date inconsistencies and verify the shipping lead-time calculation before using predictive results for business decisions.

Priority 2 — Rebuild and validate the models

Retrain the models on validated data. Use chronological validation, compare against a simple baseline, and report MAE, RMSE, and R^2. Select a model only if it demonstrates meaningful predictive value.

Priority 3 — Improve the factory-reallocation analysis

Add actual capacity, product compatibility, route distances, freight costs, and manufacturing costs. Calculate potential savings only when the required inputs are available and verified.

Priority 4 — Strengthen the dashboard

Clearly display data-quality warnings, model limitations, the assumptions behind each scenario, and the distinction between candidate assignments and validated recommendations.

Priority 5 — Conduct a controlled pilot

Test selected recommendations against actual operational outcomes before wider adoption.

9. Expected Business Value

Once the data and model limitations are resolved, the system could support:

- Better visibility into shipping patterns.
- More informed comparisons of shipping modes.
- Evidence-based evaluation of factory assignments.
- Improved coordination between production, logistics, and distribution.
- Scenario analysis for balancing cost, delivery speed, and profitability.
- More transparent decision-making for business stakeholders.

These are potential benefits, not yet verified financial outcomes. Quantified savings and performance improvements must be established through validated analysis and operational testing.

10. Conclusion

The Nassau Candy Factory Reallocation & Shipping Optimization Recommendation System establishes a foundation for data-driven shipping analysis and factory-allocation decision support.

The project demonstrates how historical data, machine learning, scenario simulation, and an interactive dashboard can be combined into a business decision-support workflow.

However, the current findings reveal two important limitations: shipping-date inconsistencies and weak predictive performance. Factory-reallocation candidates also require validation against actual capacity, compatibility, and cost constraints.

The immediate next step is to validate the underlying data, rebuild and test the predictive models, and verify the factory-reallocation scenarios before using the system to guide real operational decisions.

Report: Nassau Candy Factory Reallocation & Shipping Optimization — Executive Summary
Scope: Business findings, model performance, risks, recommendations, and next steps
Important limitation: Findings and potential benefits should not be interpreted as verified operational improvements or realized cost savings.